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Visualizing International Science and Engineering Fair Trends (2014 – 2024)

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Introduction

The first science fair in the United States was held in 1928 to encourage observation of the natural world. Sarah Scripps, a history professor at the University of Wisconsin at Stevens Point who has studied science fairs, said that, "After World War II, the fairs took on a dimension of national security. Instead of being about children's learning and general appreciation of science they became more about grooming a workforce." Projects changed drastically. Participants became competitors and earthworm dioramas became scientific claims and prototypes. Colleges and companies are seeing the value in encouraging science fairs, offering up to hundreds of thousands of dollars in special awards and internships. In Texas, Texas A&M University practically has a monopoly on science fair competitions, and Regeneron sponsors the two most prestigious global science fair competitions, the International Science and Engineering Fair and the Science Talent Search.

Science fairs are a crucial educational milestone for American students from elementary school to high school. While they can easily be just another grade, elite students take on professional-level research and compete at up to the international level. The International Science and Engineering Fair, or ISEF, is the pinnacle of this experience as the largest international pre-college science competition. ISEF sees thousands of projects annually and gives out even more in award money, opportunity, and prestige. ISEF finalists are a specific type of student and competitor, who have not only identified a research question, designed a rigorous methodology, and produced defensible findings, but who have also advanced through a gauntlet of regional and national selection. The abstracts the students produce provides various information about their demographics, competition category, and findings, and are collected by ISEF.

Yet the science a student chooses to pursue does not emerge in a vacuum. It reflects the cultural moment, personal interests and experiences, connections through mentors and resources, prestige, and what their goal might be. Understanding which scientific topics attract ambitious young researchers, and which of those topics correlate with competitive success, requires both domain knowledge and strong data visualizations to communicate interdisciplinary trends and findings from a large dataset. This study addresses the question of topic popularity over time and how this affects competitive outcomes like award amounts. This exploration will help

understand what students do, what ISEF and other science fairs reward, and what piques the public's interest. Answering these questions can help understand the importance of science fair in driving and responding to the cultural and educational moment as well as the equity of science fair judging outcomes.

Core Hypotheses

1. Artificial intelligence, machine learning, and computing topics have grown significantly as a share of ISEF submissions from 2014–2024.
2. Certain ISEF categories receive systematically larger award dollar amounts, regardless of win frequency.
3. Projects in categories that have grown in popularity over time win awards at a higher rate.

Related Works

There is limited data and research available about the environment of science fair, a clear gap compared to the depth and breadth of resources invested into science fairs. The existing literature focuses almost exclusively on the student experience for this involved in science fair, such as their motivations, attitudes, learning outcomes, and equity of access.

Current literature has established that science fair participation leads to greater understanding of procedural scientific knowledge but only modest increases in STEM interest, as these students are predisposed towards science already by participating (Schmidt & Kelter, 2017; Dionne et al., 2012). It is still debated whether science fair participation results in increased scientific understanding for the general student population (Yasar & Baker, 2003). Voluntary participation combined with adult mentorship and external resources results in more positive experiences and better outcomes for students (Grinnell et al., 2020; Czerniak, 1996).

What is absent from the literature is a qualitative analysis of the content of student research, student topics of interest over time, and whether this information correlates with success. While Korkmaz (2012) notes that topic choice does vary by gender, seeing in his subjects that male participants gravitate towards physical science and engineering while female students prefer biological and social sciences, these choices were not examined as relating to success and outcome. Czerniak (1996) recognizes an inequity in science fairs and examines

predictors of success but focuses on structural variables, such as resources and parental support, rather than the scientific content of projects itself.

This research directly addresses this gap by analyzing the abstracts, categories, content, source, and success of nearly 15,000 ISEF abstracts from 2014 to 2024, in order to uncover trends in topic/keywords and prize money distribution.

Methods

Data Collection

The dataset used is a Kaggle dataset entitled "All ISEF Projects" (Kaiser, 2022) scraped from the Society for Science's Abstract Search. The dataset contains 14,911 project entries from the years 2014 to 2024. Each unit is an individual project. The attributes of the entries in this dataset are described in **Table 1**.

Table 1: Column Titles, Data Types, and Descriptions of Dataset Attributes

Attribute Title	**Data Type**	**Description**
title	Object	Title of the project (text-type)
category	Object	What category the project participated in (text category-type)
year	Int64	What year the project competed (numerical)
schools	Object	The schools each member was from (infrequent) (text category-type)
abstract	Object	Each project's abstract (text-type)
country	Object	The country the participants were from (text category-type)

State	object	The state the participants were from (if applicable) (text category-type)
Province	Object	The province where the participants were from (if applicable) (text category-type)
awards	Object	Awards each project won (text-type)

Data Cleaning

The CSV file of the aforementioned Kaggle dataset was uploaded into Google Colab for easy understanding and manipulation with Python code.

The distribution of the "Year" attribute was checked to ensure there were not outliers from the ten-year timespan. The unique values for both the "country" and "State" attributes were checked to ensure all states and countries were valid. The "Province" element was also checked, but there is so much variation within that encoding without the requisite domain knowledge that this was left alone. Very little of the geographic information is included in the visualizations so this was deemed acceptable. The "schools" attribute was left alone as this was not included in calculations and visualizations, and, due to the time span and international nature of the competition, would be difficult to verify.

Next, duplicate detection was run to catch duplicate projects by title, abstract, country, and year to remove any artificial inflation or deflation of target metrics. A total of 73 duplicates and 38 duplicate projects were detected and removed from the dataset for analysis, while the first occurrence of the project remained. The dataset used for further analysis now has a total of 14,873 projects. A null audit was run to determine the amount of projects with missing abstract information by year, which showed a consistent approximately 20% null rate by year, with the most recent years (2021 – 2024) having a lower null rate, showing a likelihood of more extensive record-keeping. Empty "title" and "abstract" cells were filled with two sets of double quotation marks ("") rather than removed, as there can still be important information from either the title, category, or abstract to pull out topics and keywords.

Exploratory Data Analysis and Processing

Award Processing

The “awards” attribute was processed in a few ways. A function called “parse_awards” was developed to treat the column contents as text, then convert it into a list. Only the valid content was kept in the running “award_list”. Any empty content or anything that resulted in an exception being thrown returned an empty list. This function was used to create the “awards_list” column in the ISEF dataset. Another column was created, called “won” that used the “awards_list” column to check if the project had won an award and had a list containing at least one award element, and returned “True” if so, “False” otherwise. This provided a more usable format for awards and an easy indicator column for winning projects at any level without the detail of the specific award.

Next, the awards were processed to extract the monetary value of the award received. A function called “extract_award_money” was created to take the input of the awards list (“award_list”) and begin a counter of values found, starting at 0. It begins at zero so that if the function encounters an empty list, it returns a monetary value of 0. If the function encounters award elements in the list, it begins to search for a pattern: once it finds a dollar sign ($), it captures the following digits and commas. Next, it updates the total counter with the value found, removed the comma, and converts to an integer value for numerical processing. This, when applied to the ISEF dataset, creates the “award_money” column, which shows the total monetary value of awards received for each project entry. The cumulative award money, which includes ISEF-assigned prizes for place awards as well as special awards and scholarships, was then used to assign an award tier to show the magnitude of prize money rewarded. The “award_tiers” function implements a hierarchical percentile-informed bracketing approach to classify ISEF project award values into ten discrete tiers. The breakpoints are established by examining narrower slices of the distribution and checking for meaningful divisions by percentile. These bins have large variations in the frequency of their contents, such as in Tier 7, because of how ISEF awards are distributed, and it seems that the $25,000-$50,000 band is an unlikely award tier and essentially a dead zone. Tiers 8 and 9, with awards of at least $100,000, are to include the projects with top prizes (generally $75,000) as well as smaller categorical wins, and Tier 9 shows those projects that were inflated with large special awards and scholarships. Special awards have

different qualifications for judging as compared to ISEF judging. Award tiers were assigned as indicated in **Table 2**.

Table 2: Award Tier and Monetary Value

Award Tier	**Monetary Value Range (Total, $)**
None	=0
Tier 1	(0, 500]
Tier 2	(500, 1000]
Tier 3	(1000, 2000]
Tier 4	(2000, 5000]
Tier 5	(5000, 10000]
Tier 6	(10000, 25000]
Tier 7	(25000, 50000]
Tier 8	(50000, 100000]
Tier 9	>100000

Category Processing

The "category" attribute required several diagnostic and processing steps before it could be used reliably in analysis.

The first technique was a null audit, which examined the category column year by year to count how many entries were missing a value. This was important because a missing category cannot be grouped, filtered, or visualized, and a high null rate in a given year signals a systemic data problem rather than a one-off entry error. The audit revealed that 68.4% of 2015 entries and 82.3% of 2016 entries had no category value. Many of those same entries also lacked an abstract, showing an issue with the data quality from the source. Even though approximately 20% of projects from each year were missing an abstract, this combination of essential missing data could cause a severe degradation in the analysis. As a result, 2015 and 2016 were excluded from all category-level analysis and documented as a data gap.

The second technique was a category name inventory, which collected every distinct category name appearing in each year of the dataset. This was necessary because ISEF restructured its competition categories between 2016 and 2017, creating more specific categories

out of broad topics like “Computer Science” and “Medicine and Health”. This shows a total of 17 categories in 2014, a drop to 9 categories in 2015 and 2016, and consistently 21-22 categories from 2017 onward as the new system was implemented. This also further displays the issue with 2015 and 2016 projects having a severe lack of data. ISEF can update the specifics of their topics regularly, but the new system remains consistent with 20-22 categories. Without this check, old and new category names would appear to be separate topics, causing undercounts for broad legacy categories and overcounts for their successors.

The third technique was a presence matrix, which produced a table with years as rows and category names as columns, where each cell contained the number of projects in that category that year. Because ISEF runs all active categories every year, any cell containing zero is a signal for review. A subset of only categories that experienced a “zero year” (a year in which it had zero projects at ISEF) was examined and the following was determined:

1. Animal Sciences, Behavioral and Social Sciences, Biochemistry, and Biomedical Engineering were introduced in 2015.
2. 2015 and 2016 are missing a vast majority of categories, not as indicative of the restructure, but as an issue with the data source.
3. Engineering Technology: Statics and Dynamics was introduced in 2022, while Engineering Mechanics was eliminated (category renamed).
4. Energy: Chemical and Energy: Physical were eliminated in 2022, while Energy: Sustainable Materials and Design was introduced (category combination and renamed).
5. Technology Enhances the Arts is the newest category, as it was introduced in 2024.

Due to the restructure and data issues, 2014, 2015, and 2016 will not be included in further analysis, and the dataset has been updated with a total of 10,759 project entries. In the category-based dataset, there are a total of 25 categories, consistent with all category changes from 2017 onwards. To account for the category changes as well as to work with general topics, a “general_category” column was added to the dataset following the map in Table 3. This also helps account overall topics, as competitors can choose their category strategically.

Table 3: Broad Categorization Mapping

General Category Name	Competition Categories Included
Physics & Energy	• Energy: Physical • Physics & Astronomy • Energy: Chemical • Energy: Sustainable Materials and Design
Earth & Environment	• Environmental Engineering • Earth and Environmental Sciences • Plant Sciences
Engineering & Physical Sciences	• Engineering Mechanics • Embedded Systems • Materials Science • Robotics and Intelligent Machines • Biomedical Engineering • Engineering Technology: Statics & Dynamics • Technology Enhances the Arts
Life Science	• Microbiology • Computational Biology and Bioinformatics • Cellular and Molecular Biology • Chemistry • Biomedical and Health Sciences • Biochemistry • Translational Medical Science • Behavioral and Social Sciences • Animal Sciences
Computing & Math	• Systems Software • Mathematics

After adding the general categories, the workable dataset includes the following derived attributes, as described in **Table 4**.

Table 4: Column Titles, Data Types, and Descriptions of Derived Attributes

Attribute Title	**Data Type**	**Description**
text	Object	Title content + " " + Abstract content Full text content of unique project (text-type)
awards_list	Object	Store awards from "awards" as a list, empty list otherwise (text list-type)
won	Bool	True if project has won awards, False otherwise (T/F)
award_money	Int64	Total amount of money won from awards (numerical-type)
top_75	Int64	If award money won puts project in top 75% of winners (0 if not, 1 if so)
over_10k	Int64	If project won over $10,000 (0 if not, 1 if so)
over_50k	Int64	If project won over $50,000 (0 if not, 1 if so)
award_tier	Object	Tier of total award money won as described by **Table 2** (text category-type)
general_category	Object	General combined category of project as described by **Table 3** (text category-type)

Overall Summaries

There were initially 14,911 entries in the dataset, with 14,873 projects used for further analysis after duplicates removed. Of these projects, 2,778 projects (33.1% of all projects in competition years 2017-2019, 2021-2023) are classified as having won an award. 2020 was eliminated from this analysis as competition did not happen due to COVID, and 2024 awards are missing from the data due to a collection error. Overall, 90 countries are represented with a total of 25 categories having been represented across 2017 – 2024.

Table 5: Number of Projects and Number of Winners by Years

Year	Number of Projects	Number of Winners
2017	1403	456
2018	1390	460
2019	1426	458
2020	1027	N/A
2021	1452	467
2022	1400	452
2023	1316	485
2024	1345	N/A

Among winners, there is a minimum award value of $0.00, a maximum award value of $252,000, a median award value of $1,000, and a mean award value of $4,184. Most submissions, even if classified as a winner of any form of final ISEF or special award, do not receive prize money, which is the classification for award tiers.

Award Tier Distribution

The first observation is that the full distribution is extremely right-skewed and zero-heavy, as most projects do not receive any reward money, regardless of if they are classified as a winner. Among those who won award money, the top 50% of projects spans the entire range from $1,000 to the maximum of $252,000. Therefore, subsequent percentile breakdowns were used to clarify the tiers, as described in Table 2. Each tier therefore corresponds to a meaningful density cluster in the actual distribution rather than an arbitrary linear or logarithmic spacing.

Chart 1: Award Money Tier Distribution for ISEF Projects Competition Years (2017-2019, 2021-2023)

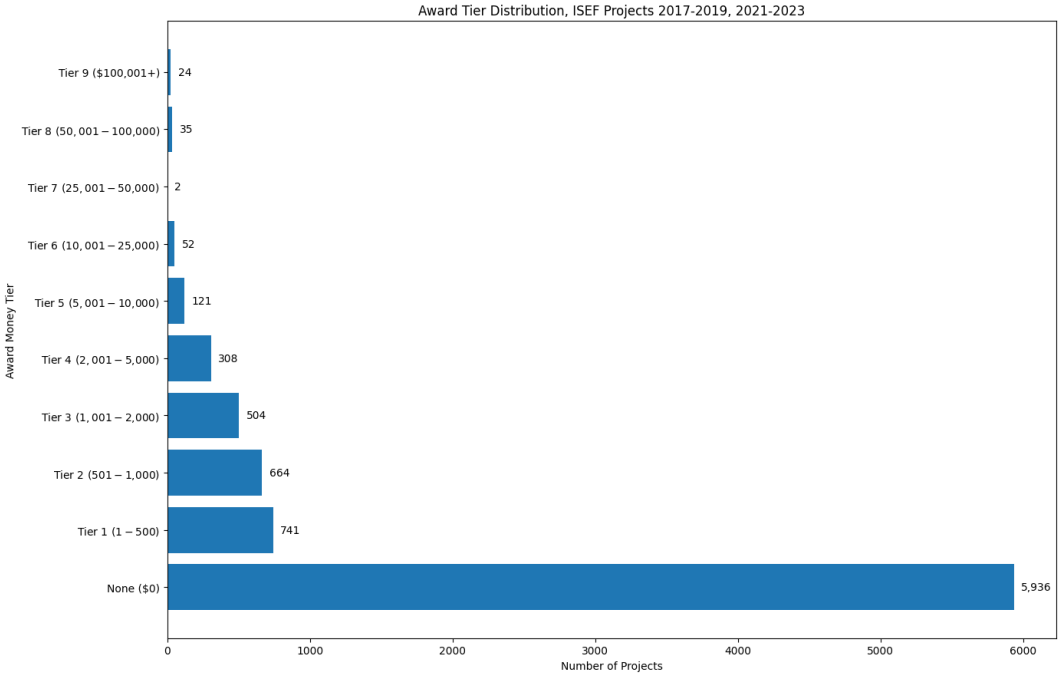
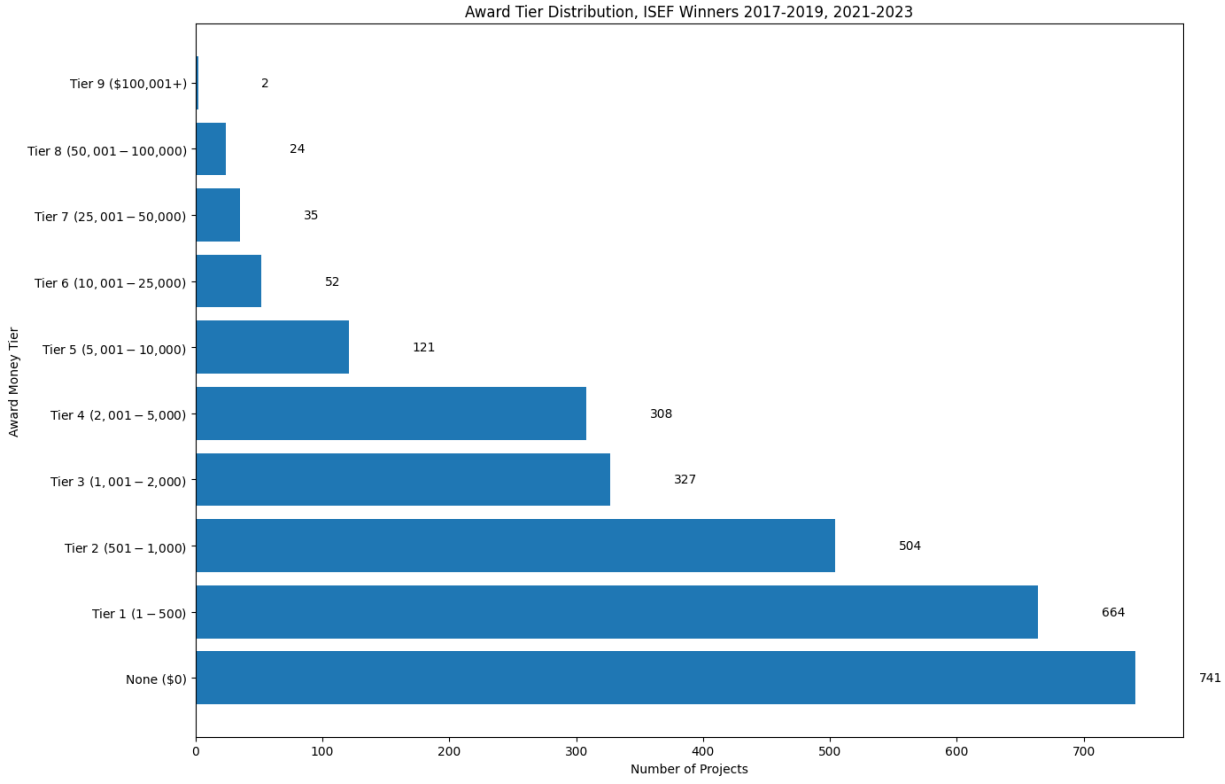


Chart 2: Award Money Tier Distribution Among ISEF Winners, Competition Years (2017-2019, 2021-2023)



These two charts present complementary views of the same award data across the competition year window of 2017–2019 and 2021–2023. The first chart shows the full population of all submitted projects while the second chart shows only projects identified as winners (received any form of recognition, regardless of monetary value).

The most immediate feature of **Chart 1** is the dominance of the None ($0) bar, representing 5,936 projects (approximately 71% of all competition year submissions). This result reflects the structural reality of ISEF, that most competitors do not walk away with money, and even those who receive an award recognition do not necessarily translate those results into monetary compensation. The sheer size of the non-winning population relative is critical for **Hypotheses 2 and 3** because it means that the decision of whether to award a project at all is a categorically different threshold from the decision of how much to award it.

Among projects that did receive monetary awards, both charts reveal a consistent and strongly right-skewed distribution. Tier 1 ($1–500) is the single most populous monetary tier with 741 projects, followed by Tier 2 ($501–1,000) with 664, and Tier 3 ($1,001–2,000) with 504. This descending pattern reflects the underlying ISEF grand award structure: Fourth Awards ($500), Third Awards ($1,000), and Second Awards ($1,500) are distributed more broadly than First Awards ($3,000) or Best of Category ($5,000), and special awards that exceed these amounts are progressively rarer. The decline from Tier 1 through Tier 4 represents the expected shape of any competitive award system where fewer prizes exist at higher levels.

A critical anomaly appears at Tier 7 ($25,001–$50,000), which contains only 2 projects in the full-population chart. This discontinuity was identified during the data audit as a natural gap in the award distribution, and both charts confirm it visually. No grand award tier lands in the $25,000–$50,000 range: the top ISEF grand prize ($75,000) exceeds it, and the Best of Category combined with a First Award totals only $8,000. Projects appearing in this range arrived there through stacking of multiple special awards. The inversion above Tier 7, where Tiers 8 and 9 are larger, reflects the scholarship stacking phenomenon, which can boost a project that did not receive as much ISEF recognition into the highest reward tier. This confirms that the upper tiers measure total compensation from all sources rather than judge-assigned placement, which has direct implications for how H5 is interpreted.

The winners-only chart **(Chart 2)** introduces a category absent from the full-population view: 741 projects classified as None ($0) despite having a non-empty awards list. These are projects that received recognition but no monetary compensation. For **Hypothesis 3**, this reinforces the argument that award magnitude and award occurrence are not just statistically independent but that they are measuring fundamentally different things.

For **Hypothesis 1**, the existence of clear tier boundaries means that if AI/ML topic prevalence is correlated with higher award tiers, that relationship is detectable against a well-defined background distribution. For **Hypothesis 2**, the distribution confirms that there is meaningful variation in award outcomes across the population. With 2,037 projects distributed across Tiers 1 through 6 in the safe analysis window, there is sufficient statistical density to compute stable win rates by category and test whether category growth predicts award tier rather than just award occurrence. For **Hypothesis 3**, the structure of **Chart 2** shows that the distribution of award dollars does not simply mirror the distribution of winning probability. A system where winning often (high win rate) and winning large (high award value) were the same phenomenon would produce a smooth, consistently decreasing distribution.

Computing Keywords Analysis

Image 1: Results Table of Computing Keywords Percentage and Change by Year

Keyword % of all projects by year:

keyword	Artificial intelligence	Computer vision	Deep learning	Machine learning	Neural network
year					
2014	0.72	0.29	0.07	0.93	0.79
2015	0.59	0.44	0.15	0.96	0.82
2016	0.36	0.22	0.44	1.75	1.02
2017	0.71	0.50	0.71	3.21	2.21
2018	1.65	0.65	1.37	4.46	4.68
2019	1.54	0.98	2.03	4.84	5.26
2020	1.75	0.88	2.73	6.62	5.74
2021	2.13	1.38	3.99	7.44	7.37
2022	1.93	1.00	3.57	8.21	7.43
2023	2.43	0.99	4.03	6.53	6.08
2024	2.68	1.19	4.68	7.66	7.14

Change from 2014 to 2024:

Machine learning	: 0.93% → 7.66%	(8× increase)
Deep learning	: 0.07% → 4.68%	(67× increase)
Neural network	: 0.79% → 7.14%	(9× increase)
Artificial intelligence	: 0.72% → 2.68%	(4× increase)
Computer vision	: 0.29% → 1.19%	(4× increase)

This table **(Image 1)** presents the percentage of all ISEF project abstracts containing each of five AI/ML-related keywords by year. This uses the entire dataset from 2014 to 2024, as the

key component is the complete text column, which is fairly full across all years. Each cell shows the proportion of that year's total submissions in which the relevant keyword appeared anywhere in the project text. The bottom section shows the overall magnitude of change across the eleven-year period.

Every single keyword increased across the eleven-year period, with minor fluctuations due to yearly submissions and data quality. However, the overall trend shows that each independently tracked term move in the same direction. Machine learning grew from 0.93% to 7.66%, an 8-fold increase. Neural network grew from 0.79% to 7.14%, a 9-fold increase. Deep learning, starting from a near-zero baseline of 0.07% in 2014, reached 4.68% by 2024, which is a 67-fold increase that reflects not just growth but the emergence of an entirely new subfield within the dataset's timeframe. Artificial intelligence and computer vision each quadrupled, growing from 0.72% to 2.68% and 0.29% to 1.19% respectively.

The most dramatic acceleration occurred from 2018 to 2021. Machine learning jumped from 3.21% in 2017 to 4.46% in 2018 and reached 7.44% by 2021, which is its highest recorded value in the dataset. Deep learning and neural network keyword presence also accelerated in a similar manner. This corresponds to when AI became a more mainstream topic and more broadly available. Afterwards, there is a bit of plateau and some slight reversal as the novelty of the technology wore off and branches out into different fields.

The data documented in this table is more than sufficient to pursue **Hypothesis 1**, which states that ISEF submissions have shifted toward AI/ML-oriented science since 2014. The large sample sizes, use of multiple percentage points, consistency across keywords, and availability of granular data for further analysis make this data acceptable to confirm the findings.

Category Growth vs. Win Rate

Image 2: Results Table of Category Growth and Win Rate

```
Pearson r (growth vs win rate): -0.001

All categories — growth vs win rate (sorted by growth):
                                category  n_projects  win_rate_pct  pct_point_growth
       Robotics and Intelligent Machines         443         37.92              2.51
Computational Biology and Bioinformatics         380         31.05              1.59
           Translational Medical Science         384         30.99              0.96
                  Physics and Astronomy         438         35.16              0.88
              Environmental Engineering         550         30.91              0.86
                        Animal Sciences         382         26.70              0.30
                           Biochemistry         235         28.94              0.23
          Behavioral and Social Sciences         390         39.49              0.22
                  Biomedical Engineering         501         32.14              0.19
                               Chemistry         423         36.41             -0.12
                             Microbiology         411         27.98             -0.13
        Earth and Environmental Sciences         562         32.74             -0.30
                        Systems Software         347         43.80             -0.30
          Biomedical and Health Sciences         487         29.57             -0.43
                           Plant Sciences         372         25.00             -0.47
          Cellular and Molecular Biology         297         31.65             -0.53
                        Embedded Systems         275         37.45             -0.61
                             Mathematics         280         44.64             -0.94
                       Materials Science         373         29.76             -1.62

Top 5 fastest-GROWING categories:
                                category  pct_point_growth  win_rate_pct
       Robotics and Intelligent Machines              2.51         37.92
Computational Biology and Bioinformatics              1.59         31.05
           Translational Medical Science              0.96         30.99
                  Physics and Astronomy              0.88         35.16
              Environmental Engineering              0.86         30.91

Top 5 fastest-SHRINKING categories:
                     category  pct_point_growth  win_rate_pct
             Materials Science             -1.62         29.76
                  Mathematics             -0.94         44.64
               Embedded Systems             -0.61         37.45
Cellular and Molecular Biology             -0.53         31.65
                Plant Sciences             -0.47         25.00

Highest win rates overall (all categories ≥100 projects):
                                category  n_projects  win_rate_pct
                             Mathematics         280         44.64
                          Systems Software         347         43.80
Energy: Sustainable Materials and Design         167         43.11
          Behavioral and Social Sciences         390         39.49
       Robotics and Intelligent Machines         443         37.92
```

This output **(Image 2)** presents the relationship between category-level submission growth and award win rates across the competition years window, along with summaries of the fastest-growing and fastest-shrinking categories and the five highest win rates among all categories with at least 100 projects.

The Pearson r value of -0.001 is near zero and shows that knowing how much a category has grown or shrunk provides essentially no correlation or information about what percentage of

that category's projects receive awards. The negative value is practically meaningless due to the magnitude and shows the data offers no support for **Hypothesis 3**, that judges reward timely or growing science at a higher rate than established or contracting science. Acorrelation of −0.001 does not mean that category membership is unrelated to win rate but that this variation in win rates is distributed essentially randomly with respect to whether a category is growing or shrinking.

The two fastest-growing categories are Robotics and Intelligent Machines (+2.51 percentage points) and Computational Biology and Bioinformatics (+1.59). These categories have win rates of 37.92% and 31.05% respectively. These are above-average rates, which might initially seem to support Hypothesis 3. However, even among the growing categories, win rates span from 26.70% to 39.49%, which is nearly the full range of the entire dataset. The shrinking side of the table is even more informative. Materials Science is the fastest-shrinking category at −1.62 percentage points, and its win rate of 29.76% is unremarkable. But Mathematics, the second-fastest shrinking at −0.94 percentage points, has the single highest win rate in the dataset at 44.64%. Embedded Systems is shrinking at −0.61 and has a 37.45% win rate, which is well above average. Cellular and Molecular Biology is shrinking and below average at 31.65%. Again, no pattern emerges. The shrinking categories are no more or less rewarded than the growing ones.

The bottom section of the output, which shows the highest win rates among categories with at least 100 projects, shows that Mathematics, Systems, Energy: Sustainable Materials and Design, Behavioral and Social Sciences, and Robotics and Intelligent Machines are the top-performing categories. These five categories span a wide range of scientific domains and growth trajectories. This finding has direct implications for the broader research question. If timely or fashionable categories do not win at higher rates, and the highest-rewarded categories are defined by disciplinary characteristics rather than cultural relevance, then the science fair's reward system appears to be operating on criteria that are largely insulated from trends in public or institutional scientific interest.

This output is entirely sufficient to pursue **Hypothesis 3**, even if resulting in no variation from the null hypothesis. The sample sizes, range of growth rates, and data clustering allows the data to perform well to these ends. The highest win rates cluster in categories characterized by

formal methodological rigor and well-defined evaluation criteria regardless of whether those categories are growing or shrinking in popularity.

Award Money Independent of Win Rate

Image 3: Results Table of Award Money Analysis Independent of Win Rate

```
Pearson r (win rate vs avg award among winners): 0.142
(Near zero = win frequency and prize size are essentially independent)

All categories — win rate vs avg award $ (sorted by avg award):
                                Category  n_projects  win_rate_pct  avg_award_winners
          Earth and Environmental Sciences         724          25.4            10382.0
                          Energy: Chemical         135          26.7             8302.0
                          Embedded Systems         347          29.7             7740.0
                 Environmental Engineering         686          24.8             7432.0
            Translational Medical Science          520          22.9             6819.0
            Behavioral and Social Sciences         494          31.2             6300.0
                     Engineering Mechanics         370          26.2             6223.0
                          Systems Software         446          34.1             5792.0
                              Biochemistry         299          22.7             5182.0
                                 Chemistry         543          28.4             4975.0
  Energy: Sustainable Materials and Design         258          27.9             4917.0
Engineering Technology: Statics & Dynamics         226          24.8             4639.0
  Computational Biology and Bioinformatics         477          24.7             4572.0
            Biomedical and Health Sciences         646          22.3             4306.0
                    Biomedical Engineering         633          25.6             3189.0
            Cellular and Molecular Biology         382          24.6             3024.0
                          Energy: Physical         100          28.0             2906.0
        Robotics and Intelligent Machines          577          29.1             2869.0
                     Physics and Astronomy         563          27.4             2583.0
                            Plant Sciences         501          18.6             2298.0
                           Animal Sciences         476          21.4             2184.0
                              Microbiology         504          22.8             2038.0
                               Mathematics         355          35.2             1859.0
                         Materials Science         476          23.3             1613.0

Highest avg award $ among winners:
                          Category  win_rate_pct  avg_award_winners
Earth and Environmental Sciences          25.4            10382.0
                Energy: Chemical          26.7             8302.0
                Embedded Systems          29.7             7740.0

Lowest avg award $ among winners:
         Category  win_rate_pct  avg_award_winners
Materials Science          23.3             1613.0
      Mathematics          35.2             1859.0
     Microbiology          22.8             2038.0

Key contrast (Mathematics vs Earth & Env. Sciences):
  Mathematics
    Win rate:        35.2%
    Avg award ($):   $1,859
  Earth and Environmental Sciences
    Win rate:        25.4%
    Avg award ($):   $10,382
```

This output **(Image 3)** presents the relationship between how often each ISEF category wins awards and how much those winning projects receive on average. The category-approved dataset was used for this analysis, so this represents 2017 – 2024.

The Pearson r of 0.142 between win rate and average award among winners is fairly negligible. It shows that approximately 98% of the variation in how much winning projects receive is due to factors outside of how often that category produces winners. While the relationship is positive, the magnitude makes the relationship nearly meaningless. For the purposes of **Hypothesis 2**, the output backs up that win frequency and prize size are essentially independent. Throughout the analysis, there are counterintuitive and counteractive examples that back up these findings.

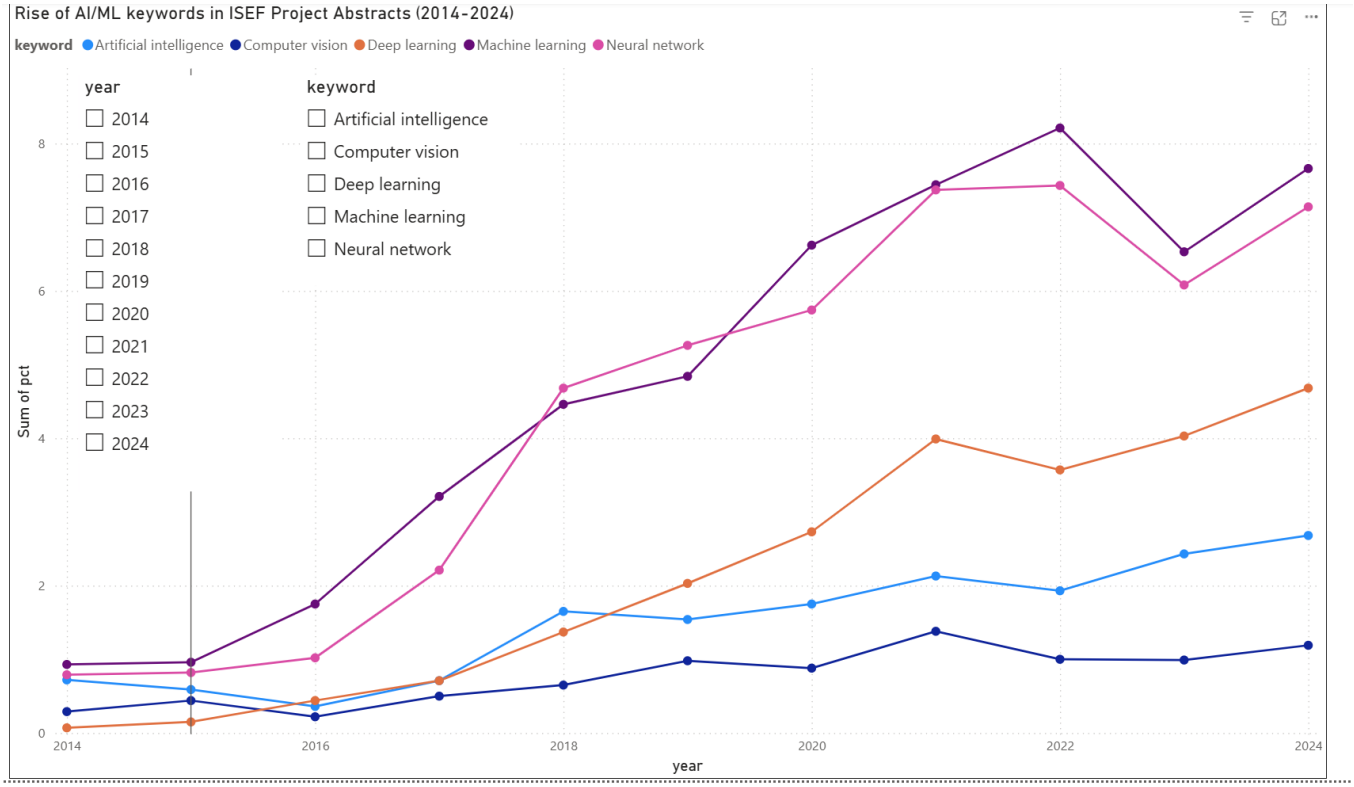
The bottom section of the output shows the most powerful findings. While Mathematics has some of the highest win rates (35.2%), it has a smaller average award amount ($1,859). Earth and Environmental Sciences, however, has a win rate of 25.4% (below median) but an average award of $10,382, nearly six times higher. These two categories sit at nearly opposite corners of the win rate versus award space. It reflects a structural feature of how ISEF's awards system is arranged. Mathematics attracts fewer special award organizations to supplement the standard ISEF category awards, while environmental research is rich in awarding projects through corporate sponsors, nonprofits, governmental agencies, and other external sponsors. The result is that a Mathematics winner is very likely to receive a grand award but unlikely to receive additional sponsorship on top of it, while an Earth and Environmental Sciences winner is less likely to be recognized at all but, when recognized, is likely to attract multiple layered awards from multiple sources. The judging system and the sponsorship system are running in parallel, and this table reveals that they are not particularly well aligned with each other.

Category awards are distributed proportionally to the number of projects entered in each category, meaning that the ratio of awards to projects is roughly held constant across categories by design, equalizing win rates. The difference is within stacking awards, receiving special awards, and moving on to grand prizes.

This data is sufficient to pursue **Hypothesis 2** due to large sample sizes, range of variation, and contrast between key categories.

Results

Chart 3: Rise of AI/ML keywords in ISEF Project Abstracts (2014-2024)



Hypothesis 1 states that ISEF submissions have shifted toward AI/ML-oriented science since 2014. This line chart **(Chart 3)** answers that question by tracking five artificial intelligence and machine learning keywords (artificial intelligence, computer vision, deep learning, machine learning, and neural networks) through their presence in ISEF titles and abstracts from 2014 to 2024. These keywords were chosen as opposed to tracking categories for two reasons: there is a large gap in category data from 2015-2016, and these keywords are related to techniques that can be applied to a project that can fall into many nontraditional categories.

Machine Learning rose from being present in 0.93% to 7.66% of all projects (an 8-fold increase), Deep Learning from 0.07% to 4.68% (a 67-fold increase), and Neural Network from 0.79% to 7.14% (a 9-fold increase). Artificial Intelligence and Computer Vision each quadrupled. The hypothesis is confirmed through this visualization. There has been a noticeable shift towards these techniques in ISEF projects as indicated by these keywords, and it is consistent across all

five independently measured terms. The visualization shows not only the end result but the growth trajectory for the reader, allowing them to track the growth phases rather than being presented with simple before-and-after numbers.

A line chart was the appropriate choice for this data for three reasons. First, the goal is to track a trend over time, where year is a continuous and ordered x-axis value. Connecting the data points via lines over the year values shows a continuous process with a relation. A line chart also allows for multiple series to be overlaid without becoming hard to read, which is essential for tracking the five keywords. It also allows for easy comparison by year of each keyword as well as overall trajectory. A line chart naturally encodes the rate of change as well, which is meaningful for tracking growth phases and trajectory.

This chart uses the vertical position of each data point on the y-axis to encode the percentage of projects containing that keyword (as indicated by color) in that year (as indicated by position along x-axis). Position along a common scale is an accurate, easy to understand visual encoding element. The x-axis encodes time following the standard Western reading conventions, with the earliest time period (2014) on the left and the latest time period (2024) on the right. This allows the user to understand the data while following their natural eye movement for the chronological progression. Since the y-axis encodes percentage of all projects containing the keyword, it begins with 0% at the bottom to the maximum value at the top. Beginning with 0% improves accuracy by putting all numbers on a common, proportional scale. The slope, which is generated implicitly by the line chart, helps display rate of change using the comparative steepness of the line.

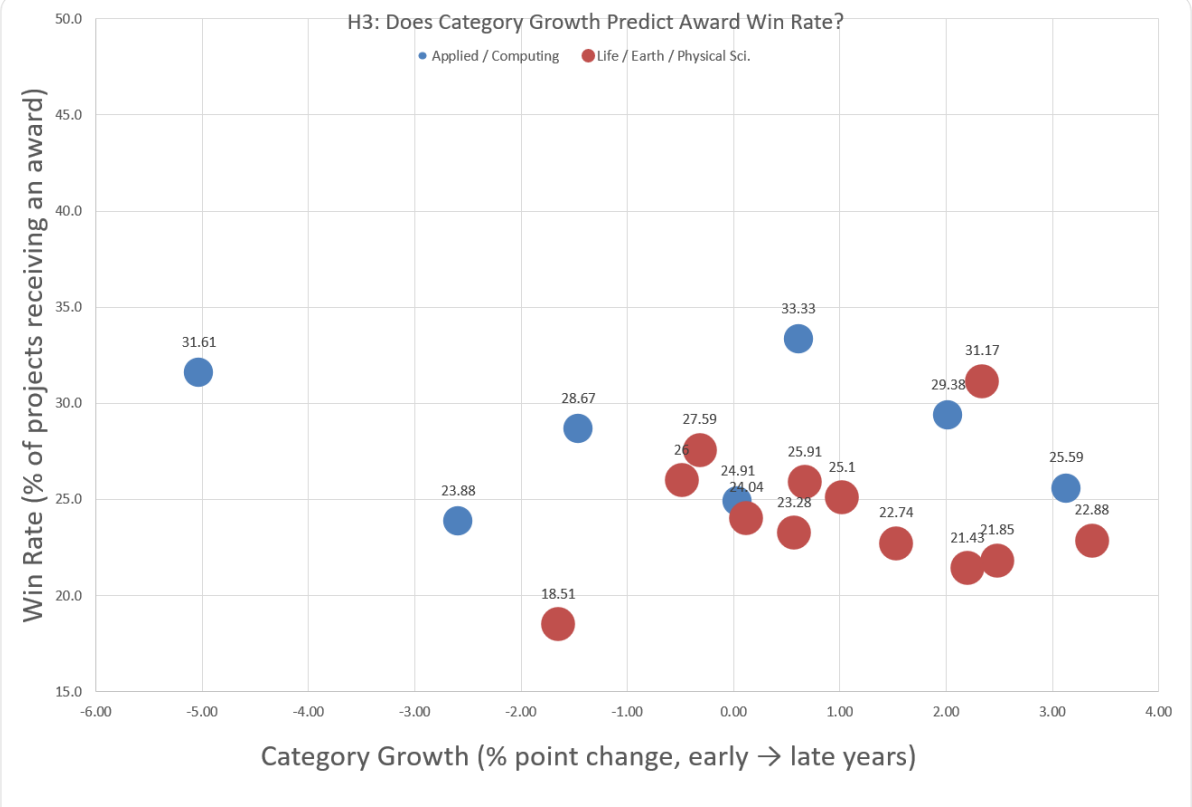
The second crucial element to understanding is color, which is used to distinguish the five keyword lines from each other. While color is harder to determine for quantitative information, color here serves as a categorical encoding. The legend provides this information for the user. Power BI's default categorical palette is used for the color choices for the keywords. There are no general colors corresponding to the keywords, so this was not incorporated. This is a qualitative palette since it is representing categories with no natural sequencing or hierarchy. The blue colors can be too similar for people with visual impairments and the reliance on color is a challenge for those with visual impairments as well, so adding a shape in addition to color for

each keyword would be helpful. Additionally, the sliders are incorporated to help with pulling out this information, but it lacks the comparative and automatic perception element.

The background of the visualization is white with light grey gridlines to help the eye track distinct data points without providing visual clutter. The blank slate helps emphasize the colors of the lines.

There are two slicers included for interactive elements for the user. The user can select a subset of keywords and/or a subset of years to focus on. They are places on the upper left-hand side of the chart, filling in the blank space created by the shape of the line chart. While this is compact, it might be confusing to not have the slicers clearly labeled and connected to the characteristics they are meant to control.

Chart 4: Does Category Growth Predict Award Win Rate?



Hypothesis 3 states that categories which have grown in popularity over time win awards at a higher rate. This is based in the assumption being that judges reward timely or culturally relevant science. This scatter plot **(Chart 4)** answers that question by placing every ISEF category simultaneously in a two-dimensional space defined by exactly those two variables. The

lack of any discernible trend answers the question for the viewer. The null hypothesis is thus proven correct. Otherwise, there would be a cluster of high-growth, high-success categories, and a cluster of low-success, low-growth categories.

The specific values reinforce this. The leftmost dot, which is the fastest-shrinking category, has one of the highest win rates. Meanwhile several growing categories (clustered between x = 0 and x = 3.5) show win rates between 21% and 26%, below the chart median. If any slight trend appears, it shows that growing categories if anything win marginally less often than shrinking ones. This can be due to larger categories being only able to reward a certain number of participants so the percentage of projects rewarded decreases.

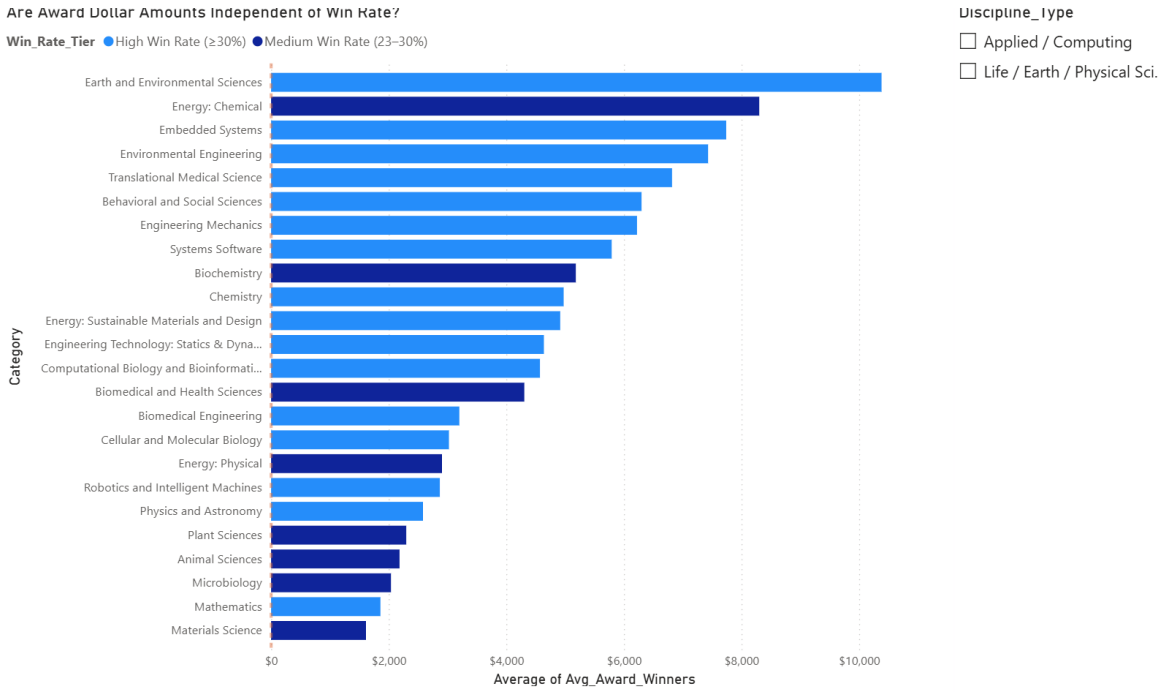
A scatter plot was chosen due to the two variables that are being plotted: two continuous, quantitative variables of category change over time and category win rate. The hypothesis asks about the relation of the two variables, so allowing the point to represent the combination of the two variables makes the relational trends and cause/effect more apparent. A scatter plot is also fairly standard fare for a correlation display.

This chart uses the position along two common scales to show the relation between category growth (x-axis) and win rate (y-axis). Position along a common scale is an accurate, easy to understand visual encoding element. Category growth is measured as the change in percentage point share of all ISEF submissions from the early years (2014 – 2017) to the late years (2021 – 2024), ranging from approximately −6 to +4 in this chart. Growth is placed on the horizontal axis because it is the predictor variable. The axis crosses zero at the left edge rather than in the center, which means negative-growth (shrinking) categories appear in the left portion of the chart and positive-growth (growing) categories in the right. This is a useful layout because it visually separates the two groups and allows a reader to assess whether the left portion of the chart looks systematically different from the right. However, there is no visual difference in the vertical range of both horizontal halves, which refuted **Hypothesis 3**. The y-axis (vertical) encodes win rate, running from 15% at the bottom to 50% at the top. The axis does not begin at zero, which is intentional as no ISEF category has a win rate near zero due to the fact that each category presented in a competition year is guaranteed a winner. If this chart started at zero, it would artificially inflate the representation of the win rates.

The second visual encoding is through color, which sets Applied/Computing categories as blue, separate from Life/Earth/Physical Science categories (red/brown). These colors are visually distinct and useful for categorical differentiation. This allows the user to look at the relation between growth and win rate through general scientific disciplines, especially as this project looks at advancements in physical and computing sciences. These disciplines, however, occupy the chart with no discernible pattern, additionally refuting **Hypothesis 3**.

The data labels on each dot show the win rate, though this is not an essential element of have the specifics of due to the limited range displayed. A more useful metric would be the specific categories labelling the data points, not just the general discipline through color. The viewer is not able to pull out the Mathematics point, for example, without previous knowledge. The visualization would be strengthened by making this change, adding the correlation coefficient, and a greater visual element at growth rate = 0 to help distinguish the categories that are shrinking versus those that are growing.

Chart 5: Are Award Dollar Amounts Independent of Win Rate?



Hypothesis 2 posits that certain ISEF categories receive systematically larger award dollar amounts regardless of how often they win. This would mean that win frequency and prize magnitude are independent dimensions of competitive success. This bar chart **(Chart 5)** answers

that question by sorting all 24 categories from highest to lowest average award value and encoding win rate tier through bar color. This includes the competition years' data and is missing the category introduced in 2024, as this year is both missing competition data and this category does not meet the 100-project minimum for analysis.

If **Hypothesis 2** were false — if high win rates and high awards traveled together — the chart would show a clean color gradient from light blue at the top to dark blue at the bottom. Instead, the two colors alternate with no discernible top-to-bottom pattern, which is the visual confirmation of independence. The two most rhetorically powerful data points are immediately readable from the chart. Earth and Environmental Sciences occupies rank 1 with the longest bar at $10,382 average award among winners, yet it only has a medium win rate. Mathematics sits at rank 23 with one of the shortest bars at $1,859, yet has a high win rate. This is consistent in **Chart 4** as well. The category that wins most often earns the least per win. The category that earns the most per win does not win especially often. This inversion is not an edge case, is the structural story **Hypothesis 2** predicts, made visible in two bars the reader's eye naturally finds because they are the extremes of the ranking.

A sorted horizontal bar chart was chosen to compare category-level differences in award magnitude based on the category's overall average award prize delivered. Individual category identification is easy with the bar chart, and the horizontal bar chart allows the user to follow the way they naturally read. Sorting by award value and encoding win rate as color creates the most direct possible visual test of the hypothesis, allowing the reader to verify there is no trend visually through both bar length and color value.

This chart uses the length along a common scale as the primary encoding, setting the average award value as the length of the bars. Length along a common scale is an accurate, easy to understand visual encoding element. The dollar-formatted x-axis allows for more precise reading while the general shape allows for trend analysis and general comparison. The x-axis starts at $0 since this is the majority of participants' experience and to not artificially inflate differences in winnings. The y-axis shows the categories, sorted from highest award value to lowest average award value. The two key variables of the hypothesis are separated to make independence clear.

The second visual encoding is through color, which sets high win rate categories as light blue while medium win rate categories are dark blue. This allows the user to see the main argument of the chart. This is missing a low win rate color and different colors or texture could be used to enhance accessibility. The current chart also shows only two tiers rather than three because no categories in this dataset fall into the Low win rate (<23%) tier during the award-safe analysis window, which is itself a finding worth noting — all 24 categories achieved at least 23% win rates during 2017–2019 and 2021–2023.

The Discipline_Type slicer on the right allows the reader to filter the chart to Applied/Computing or Life/Earth/Physical Science categories independently to analyze differences between the two domains, which could perform differently come changes in techniques. The user can also scroll over the chart to see the average award amount, total number of projects, total amount of money awarded, and discipline categorization. This information gives the user more specific information and allows them to compare the "importance" of each ranking due to the number of projects and the calculation methods. The labels of the axes and drill-down characteristics are not entirely clear to the user and more information about the chart, organization, and data included would make this a more robust visualization.

Discussion

This study set out to examine the extent to which patterns of topic popularity within ISEF project abstracts align with competitive outcomes, as well as degrees of independence and patterns in award rates. Three hypotheses were developed and tested using a combination of computational text analysis and statistical methods applied to 14,873 deduplicated ISEF project abstracts spanning 2014 to 2024. Student research has demonstrably shifted toward computing and AI-oriented science over the past decade, but this shift carries no measurable competitive advantage, and the financial rewards of winning are distributed independently of category win rates.

Hypothesis 1, that ISEF submissions have shifted toward applied and computing-oriented science since 2014, driven particularly by the rise of AI/ML terminology, was strongly confirmed across all five tracked keywords. Machine learning grew from 0.93% to 7.66% of all project abstracts between 2014 and 2024, a nearly eightfold increase. Deep learning displayed a

67-fold increase, showing not only a topic's growth bu the emergence of a new subfield. Neural network grew ninefold, from 0.79% to 7.14%. Artificial intelligence and computer vision each approximately quadrupled. Year by year, the shift accelerated as these topics became commonly available, then settled as the novelty wore off, but still maintained a strong presence and overall shift in ISEF content.

These findings are consistent with the broader literature on science fair topic selection. Korkmaz (2012) documented that student project choices reflect personal interests, parental influences, and cultural context, while Dionne et al. (2012) found that genuine content interest was the primary motivator among elite competitors at the Canada-Wide Science Fair. The AI/ML surge documented here likely reflects both genuine student interest in these fields and the increasing availability of mentors, resources, and accessible tools that lower the barrier to conducting AI-based research at the high school level.

Hypothesis 2, that categories growing in popularity would win awards at higher rates, reflecting a tendency for judges to reward timely science, was clearly refuted by the data. If anything, the slight negative shift suggests the opposite: the fastest-growing categories show modestly lower win rates than the fastest-shrinking ones. This is likely due to ISEF's reward structure rather than judge behavior. ISEF judges appear to be rewarding depth and rigor rather than topical currency. This finding is broadly consistent with the conclusions of Czerniak (1996), who identified methodological sophistication and access to professional resources as the primary predictors of science fair success, and with Grinnell et al. (2020), who found that student researchers who did more background research and demonstrated greater perseverance were more likely to win awards. The implication is that ISEF judging, at least at the category level, is not responsive to fashion, which is important for students' choice in their topics.

Hypothesis 3 posits that certain categories receive systematically larger award dollar amounts regardless of how often they win. This hypothesis was confirmed. The key contrast illustrating this independence is between Mathematics (high win, low reward) and Earth and Environmental Sciences (low win, high reward). This pattern is best explained by ISEF's reward structure, which has performance awards set out by ISEF judges as well as special awards which can reward any number of topics or behaviors. The science fair system may inadvertently

channel financial opportunity according to the interests of its sponsors rather than the merit of its competitors.

Future Work

This study represents a first systematic examination of topic trends and award structure in ISEF abstract data at scale. Several extensions would substantially deepen and broaden its findings, as follows:

1. Google Trends integration to show how topics follow or lag public interest
2. Country and university rankings to show educational equity and research topics
3. Analyzing specific ISEF placements
4. Geographic and equity analysis
5. Topics with most AI-powered tools
6. Research from more recent years
7. Latent topic modeling to uncover hidden themes

Error Analysis

There are various sources of error and data limitation in this study, as follows:

1. Documented data gap in 2015-2016 from source
2. ISEF 2020 cancellation due to COVID-19 leaves lack in award findings
3. 2024 award data gaps found in source data
4. Category taxonomy restructure – while the broad category attempted to account for this, there are likely unknown effects
5. Conflation of ISEF awards with special awards in award tier and award money calculations
6. Selection bias in ISEF population, as participants must already pass through various levels of competition

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